

Solution Requirements

Date	27 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in to access crop recommendation services and save prediction history.	High
2	Authorization levels	Farmers and administrators have different access permissions for managing recommendations and datasets.	High
3	External interfaces	Connects with agricultural datasets, weather APIs, soil databases, and the trained Machine Learning model.	High

4	Transactions processing	The system processes user inputs (N, P, K, temperature, humidity, pH, rainfall) and generates crop recommendations instantly.	High
5	Reporting	Displays crop prediction results, confidence score, soil analysis, and recommendation history.	Medium
6	Business rules, etc.	Crop recommendations are generated only after validating all required soil and climate parameters.	High
7	Compliance to laws or regulations	User information is securely stored and handled according to applicable data privacy standards.	Medium
8	<i>Other</i>	Future support for fertilizer recommendation, disease prediction, and multilingual access.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Crop recommendations should be generated quickly after user input.	Response time < 3 seconds

2	Scalability	The system should support increasing users and larger agricultural datasets.	Supports thousands of users
3	Security & Data Privacy	User data and prediction history must be securely stored and protected.	Encrypted data storage
4	Reliability & Availability	The application should remain available with minimal downtime.	99.9% uptime
5	Usability & Accessibility	Simple, responsive, and user-friendly interface for farmers on desktop and mobile devices.	Easy navigation and responsive UI
6	<i>Other</i>	Maintainable, modular, portable, and easy to upgrade with future ML models.	Modular architecture